MoneyMobile

Platform Documentation

Goal, vision, architecture, security and design of the installment phone-shop platform: web administration and Android device control.

WEB PLATFORM	ANDROID APP	DATE
Next.js 14 · Supabase	Version 1.10.1 (build 20)	3 November 2026

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1 · SUMMARY

The platform at a glance

MoneyMobile lets phone shops sell smartphones on monthly installments with confidence. The shop runs its whole installment business from one website, and every phone it sells carries the MoneyMobile app, which protects the shop until the phone is fully paid.

Moneymobile WEB: the shop's control centre

A secure, multi-shop web application for customers, guarantors, inventory, installment plans, payments, reports, employees and phone control. Hosted on Vercel, with Supabase (PostgreSQL, authentication and file storage) behind it.

Money mobile: the Android device app

A Kotlin app installed as **Device Owner**, the highest management level Android offers. It shows the customer their balance, and it locks the phone when a payment is overdue, the SIM is removed or the shop presses Lock. It unlocks automatically when the customer pays.

3
management roles: super admin, shop owner, employee

~60 s
phone check-in interval, with instant push on top

0
internet needed for the phone to lock on the due date

Key capabilities

- **One-click phone setup** from Windows or macOS (USB), or with a QR code. No typing on the phone.
- **Automatic overdue lock** every month, online or offline, with reminders 48 and 24 hours before.
- **Instant unlock on payment**: recording a payment clears the overdue months and wakes the phone by push.
- **SIM protection** for single- and dual-SIM phones: removing a registered SIM locks the phone at once.
- **30-minute offline unlock code** the shop can read out over the phone, valid with no internet.
- **Per-shop reset protection**: after a factory reset the phone asks for the shop's Google account.
- **Online app updates** published by the super admin and installed silently with one click from the shop.
- **Clean release** when the phone is fully paid: every restriction is removed and the customer owns it outright.

2 · GOAL AND VISION

Why MoneyMobile exists

Selling phones on installments helps customers own a smartphone they could not pay for at once. For the shop, the risk is a customer who stops paying and keeps the phone. MoneyMobile removes that risk, so shops can say yes to more customers.

Project goal

For the shop

Run the full installment business (selling, collecting, following up, releasing) from one place, with a phone that enforces the agreement automatically and fairly.

For the customer

Use the phone normally while paying on time, always see what is owed and when, and get clear warnings before anything happens. When the last installment is paid, the phone is completely theirs.

Vision

MoneyMobile is built as a **product sold to many phone shops**, not a tool for one shop. The platform owner (super admin) creates shops, gives each one a number of licenses (one license per phone sold), publishes app updates, and watches the health of every shop from one dashboard.

Multi-shop SaaS

Every shop's data is separated at the database level. Shops manage their own staff, customers, phones and reset-protection account.

Built for Sri Lanka

Rupee amounts, Sri Lanka time (UTC+5:30) for every due date, and support for WhatsApp-only data packages, where the phone may be offline for days.

Fair by design

Locks follow clear rules: a grace period, advance reminders, calls and SMS that keep working while locked, and automatic unlock the moment the customer pays.

> **In one sentence:** MoneyMobile makes installment selling as safe for the shop as a cash sale, and as simple for the customer as owning the phone outright.

3 · WHO USES IT

Four kinds of user, one experience

Each person sees only what they need. Permissions are enforced on the server and in the database, not only hidden in the screen.

USER	WHAT THEY DO	WHAT THEY CAN SEE
Super admin Platform owner	Creates shops and shop owners, sets license limits, resets passwords, uploads and publishes app updates, reviews audit logs and server health.	Every shop, every phone and every record on the platform.
Shop owner shop_admin	Runs the shop: employees, customers, guarantors, inventory, installment plans, payments, device management, unlock codes, reports, reset protection, "Update all" phones.	Everything in their own shop, nothing from other shops.
Employee employee	Registers customers and phones, records payments, sets phones up, locks and unlocks, handles unlock requests, updates a phone's app.	Their shop's records, limited to the customers and phones they created or were assigned.
Customer Phone user	Uses the phone, sees their balance and next due date in the MoneyMobile app, gets reminders, calls or texts the shop, enters an unlock code.	Only their own agreement, on their own phone.

4 · WHAT WE HAVE BUILT

Feature overview

Three areas work together: running the shop, controlling the phone, and managing the platform.

Shop operations (web)

AREA	WHAT IT DOES
Customers and guarantors	Customer and guarantor records with NIC (ID card) photos, front and back, stored in a private storage bucket and shown through short-lived signed links.
Inventory and phone models	Phone models, stock and individual devices with IMEI, linked to customers and agreements.
Installment calculator	Flat or reducing-balance interest at a monthly rate, down payment and a fixed software fee (no interest on the fee). All money math runs on the server.
Payment schedule	Monthly installments with an exact due date and time in Sri Lanka time, a grace period, and statuses: paid, partial, due, upcoming, overdue.
Payments	Each payment is spread over the oldest unpaid months first, saved with the phone's model and IMEI for history, and immediately re-checks the lock.
Employees and licenses	Shop owners manage their employees. Each shop has a license limit (one license per phone sold), enforced in the app and by a database trigger.
Reports and dashboards	Financed, collected and outstanding totals, active and completed agreements, and the largest outstanding balances.

Phone control (web + Android)

AREA	WHAT IT DOES
One-click setup	A setup kit for Windows and macOS, pre-filled with the shop's details: install, make Device Owner and register in one double-click. QR-code provisioning on the phone's welcome screen works with no computer.
Lock, unlock, release	Staff buttons with live status (queued → delivered → applied). Release hands the phone over permanently once the balance is zero.
Kiosk lock screen	A full-screen lock the customer cannot leave. It shows the balance, overdue amount and next due date, with buttons to call or SMS the shop, join Wi-Fi, enter an unlock code or request an unlock.
Automatic overdue lock	Locks when a month is unpaid after its due date plus grace days, every month, and also with no internet. Reminders appear 48 and 24 hours before.
SIM watch	Every SIM present at setup is registered (dual-SIM aware). Removing or swapping any of them locks the phone and raises an approval alert for staff.
Offline unlock code	A 6-digit code the shop generates and reads to the customer. It works with no internet and opens the phone for 30 minutes.

AREA	WHAT IT DOES
Location and status	Last known location, battery, network, SIM operator, app version, Device Owner status and bootloader state for each phone.
Reset protection	Each shop's own Google account is stored on its phones, so after a factory reset the phone only sets up with the shop's account. A badge shows whether each phone is protected.
Online app updates	New app versions are installed silently over the internet when staff press Update. Progress and results show on the website.

Platform management (super admin)

Shops and shop owners with edit, suspend and delete, password resets, license limits, platform-wide lists of employees, customers, devices, installments, payments and unlocks, audit logs, server health checks, and the **App Updates** console: upload → test in one shop → publish to all → withdraw.

5 · HOW IT WORKS

End-to-end flows

The server is always the source of truth. The phone checks in, receives the state it should be in, and applies it. When it cannot reach the server, it follows the last deadlines it was given.

5.1 Selling a phone

1. An employee creates the customer (with NIC photos) and, if needed, a guarantor.
2. They create the installment agreement. The server calculates the plan and builds the monthly schedule. The shop's license count is checked.
3. The phone is set up with the one-click setup kit (USB) or a QR code. MoneyMobile becomes Device Owner and registers itself to the shop with the shop's secret.
4. The server issues the phone its own **device token** and **signing key**. Every SIM in the phone is registered.
5. On its first check-in the phone receives its agreement, payment deadlines and reset-protection account. The customer now sees their balance and next due date.

5.2 The check-in (heartbeat)

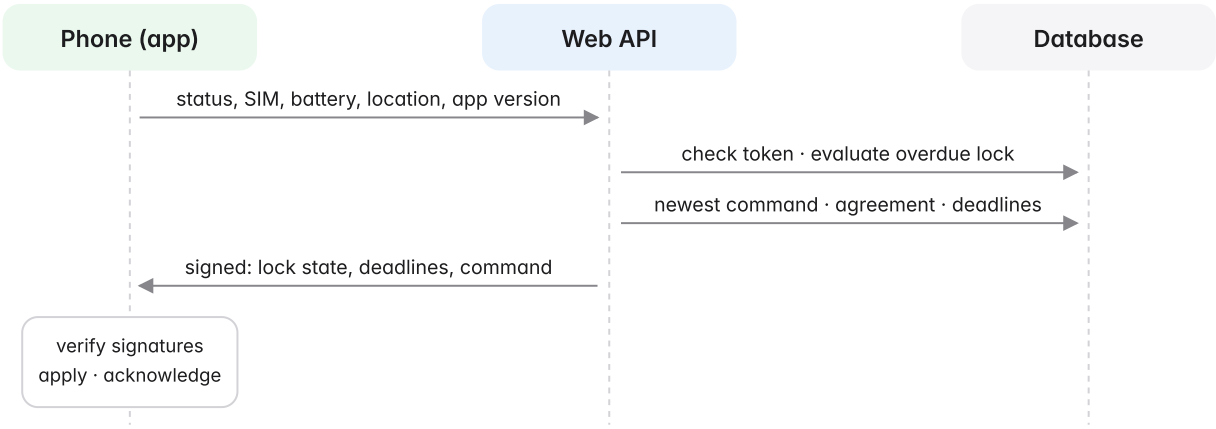


Figure 1: Every ~60 seconds (and at once when a network appears or a push arrives) the phone checks in and reconciles to the server's state.

5.3 Overdue lock, online and offline

1. On each check-in the server computes the exact moment the first unpaid month becomes overdue (due time + grace days) and the deadlines of the following months. It signs them and sends them with its own clock time.
2. The phone stores them. A deadline alarm, a 15-second watcher and every reboot check whether a deadline has passed.
3. Time is measured on a **trusted clock**: server time at the last check-in plus the phone's uptime since then. Changing the date does not help, and the date settings are locked.
4. When the deadline passes, the phone locks itself even in airplane mode. The next check-in confirms or corrects it.

5.4 Payment and unlock

Staff record the payment → it is applied to the oldest unpaid months → the server re-evaluates the lock → if every overdue month is cleared, a push wakes the phone, which unlocks within seconds. A payment never lifts a **SIM** or **manual** lock. Those carry their own reason and need staff to clear them.

5.5 SIM removed

The phone checks its SIMs every 5 seconds and on every SIM broadcast. A missing or swapped registered SIM locks the phone immediately, even offline. The alert is delivered to the server before the next heartbeat, the agreement is marked restricted with reason `sim`, and staff **Approve** (unlock) or **Reject** it. Putting the original SIM back clears the phone's own alert.

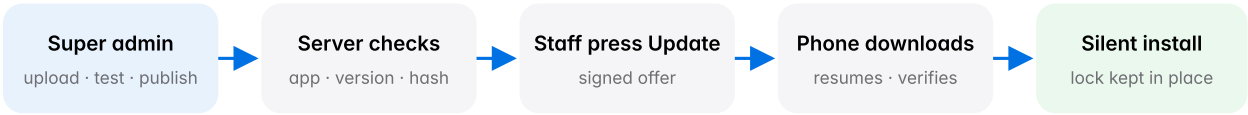
5.6 Unlock without internet

The customer calls the shop. Staff generate a 6-digit code and read it out. The phone checks the code itself, with no network, using a secret it already shares with the server. It opens for 30 minutes. If the installment is still unpaid after that, the phone locks again, online or offline, and the server keeps the same 30-minute window.

5.7 Reset protection

The shop owner enters the shop's Google account ID in Shop Settings. Every phone of that shop receives it, signed, on its next check-in, and Android stores it in a protected area that survives a factory reset. After a reset with the hardware buttons, the phone only finishes setup with the shop's account, so the customer has to return to the shop. Each phone reports whether it really holds the protection, shown as a badge.

5.8 Online app update



Android installs atomically: the phone runs either the complete new version or the unchanged old one.

Figure 2: Release pipeline from upload to phone. The published version is also what the setup kit hands out to new phones.

5.9 Release (fully paid)

1. The balance reaches zero and staff press **Release device**.
2. The phone first removes the shop's reset-protection account and confirms Android really removed it.
3. It lifts every restriction, allows uninstall and factory reset again, and gives up Device Owner as the very last step.
4. It confirms to the server, which marks the phone released and revokes its token. The customer now owns a normal phone.

6 · SYSTEM ARCHITECTURE

How the pieces fit together

A serverless web application and a managed database serve two kinds of client: staff in a browser, and phones running the MoneyMobile app.

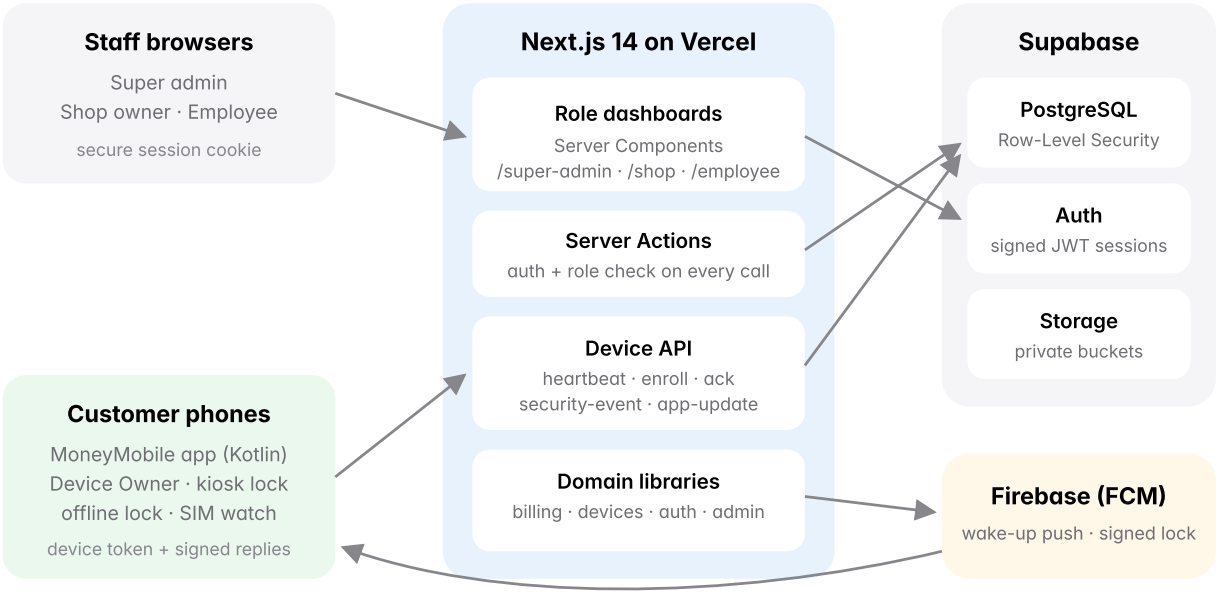


Figure 3: System overview. Staff use the role dashboards; phones use the Device API. Both reach the same PostgreSQL database, and push messages travel through Firebase.

Main components

COMPONENT	RESPONSIBILITY
Role dashboards	`app/super-admin`, `app/shop`, `app/employee`, each with an authentication gate (`layout.tsx`), loading skeletons, pages and server actions.
Device API	`app/api/devices/*`: the endpoints phones call. Authenticated with the phone's device token, not a login.
Setup-kit API	`app/api/setup-kit/*`: per-shop install kit (zip with config), public APK download and QR configuration.
Domain libraries	`lib/billing` (interest, schedule, overdue lock, totals), `lib/devices` (signing, commands, push, offline unlock, app updates, reset protection), `lib/auth`, `lib/admin`, `lib/shared` (Sri Lanka time, flash messages, storage, zip).
Android app modules	`admin` (Device Owner policies), `service` (management service, heartbeat, push), `monitor` (boot, SIM, alarms, app-updated), `update` (online updater), `provisioning` (QR and auto-enroll), `ui` (splash, enrollment, dashboard, lock screen), `util` (signatures, offline lock, unlock codes, reminders, Wi-Fi, location, secure storage).

Core data model

`shops` · `profiles` (role + shop) · `customers` · `guarantors` · `phone_models` · `devices` · `installment_agreements` (restriction + reason, balance, grace) · `installment_schedules` (monthly rows, due time) · `payment_transactions` · `device_enrollments` (the phone's live record) · `device_commands` (queue) · `device_security_events` (SIM alerts) · `device_locations` · `temporary_unlock_keys` · `app_releases` · `device_app_updates` · `audit_logs`. Every shop-owned table carries a `shop_id`, which the security policies use to keep shops apart.

How the code is organised

Architectural styles

STYLE	HOW MONEYMOBILE APPLIES IT
Client–server, multi-tenant SaaS	One platform serves many shops from a shared database. Tenants are separated by `shop_id` and enforced with PostgreSQL Row-Level Security.
Serverless backend-for-frontend	Next.js Server Components read data on the server; Server Actions perform every change. No business logic or secret keys reach the browser.
Modular monolith, feature folders	One deployable web app organised by feature (`billing` , `devices` , `auth` , `admin`), with routes grouped by role. Simple to deploy, clear to maintain.
Desired-state reconciliation	The server holds the truth (locked or not, deadlines, commands). Phones pull it on every check-in and converge to it, as Kubernetes and MDM agents do.
Hybrid pull + push	Regular polling guarantees delivery; Firebase push makes it instant. Either alone is enough for the phone to reach the right state.
Offline-first edge agent	The Android app enforces the agreement with locally stored, server-signed deadlines and a trusted clock, so enforcement does not depend on connectivity.
Defense in depth	Security is layered: database policies, server checks, signed messages, Device Owner restrictions and Android's reset protection.

Design patterns in the code

PATTERN	WHERE	WHY
Command queue + acknowledgement	`device_commands` , `queueCommand` , `/command-ack`	Lock, unlock and release are queued, delivered once, confirmed by the phone and shown live to staff.
Supersede (latest wins)	Heartbeat command delivery	Repeated clicks never build a backlog; an old Lock can never return after a newer Unlock.
Idempotency and replay guard	Last command id, FCM command id, code time steps	A repeated or replayed message is never executed twice.
Facade	`DevicePolicyController`	One class wraps every Android Device Owner call, so lock and unlock behave the same from any path.
Strategy	`calcInstallment` (flat / reducing)	Interest methods are interchangeable behind one function.
Repository	`EnrollmentRepository` , Supabase clients	Data access is separated from screens and business rules.

PATTERN	WHERE	WHY
Singleton with lazy cache	`ApiClient`, `AppPreferences`	One HTTP client and one encrypted store per process, so the lock screen stays fast.
Observer	Boot, SIM, package-replaced receivers, network callback	The app reacts to system events immediately instead of waiting.
Background worker with backoff	WorkManager heartbeat, auto-enroll, app updater	Work survives restarts and retries with exponential backoff.
State machine	App update and command statuses	Clear, finite states (pending → sent → downloading → installing → installed / failed) that can never go backwards.
Single-flight (mutex)	Heartbeat lock in `ManagementService`	Only one check-in runs at a time, even when several triggers fire at once.
Graceful degradation	Fallback queries for new columns	A database update that hasn't been run yet slows a feature down; it never breaks a page or a check-in.
Tolerant reader	Gson adapter: unknown command → `NONE`	Older apps safely ignore anything newer servers send.

8 · SECURITY

How MoneyMobile is secured

Every layer assumes the layer around it could be attacked: the browser, the network, the phone and even a forged server reply.

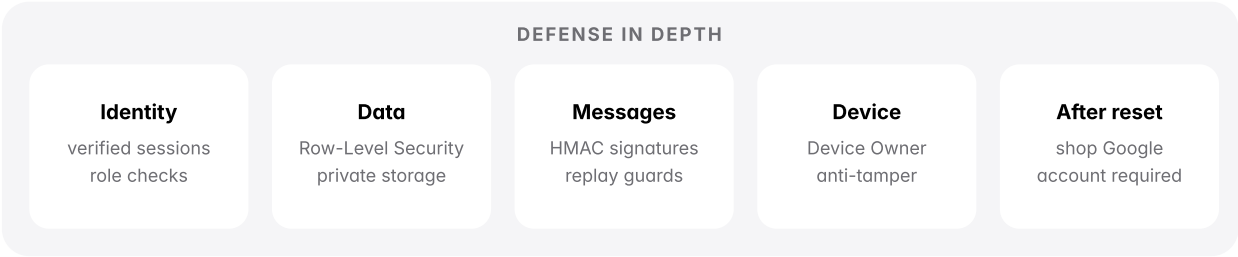


Figure 4: Five layers of protection, from the staff login down to the phone's firmware-level reset protection.

8.1 Identity and access (web)

- **Verified sessions.** Supabase Auth issues signed JWTs. `getSessionProfile()` requires the token's signed `sub` to match the user, and loads the profile through the database with that token, which rejects a forged or expired one. Disabled accounts are treated as signed out on every request.
- **Authorisation on the server.** Every Server Action checks the signed-in role and shop before doing anything. Layout gates redirect anyone who opens another role's pages.
- **Row-Level Security.** PostgreSQL policies (`is_super_admin()` , `can_access_shop()`) decide which rows each user can read or change. Employees are limited further to the customers and phones they created or were assigned.
- **Service key kept server-side.** The privileged database key is used only in server code marked `server-only` , never shipped to the browser.
- **Audit trail.** Sensitive actions (commands, payments, unlocks, staff changes, setup-kit downloads) are written to `audit_logs` .

8.2 Phone identity and message integrity

- **Device token, stored only as a hash.** Each phone gets a random token at registration. The database keeps only its SHA-256 hash, so a database leak cannot be used to impersonate a phone.
- **Signed server replies.** Each phone holds its own signing key, derived on the server with HMAC-SHA256 from a server-only secret and handed to the phone once. Commands, lock state, payment deadlines, app-update offers and the reset-protection account list are all signed. The phone ignores anything forged or altered.
- **Freshness and replay protection.** A signed lock state more than 2 hours away from the phone's trusted clock is rejected. Command ids already executed are never run again. Push commands must be signed, under 10 minutes old and not seen before.
- **Constant-time comparisons** for every signature check, so timing gives nothing away.

8.3 Offline unlock code

A 6-digit time-based code, built like an authenticator app: HMAC-SHA256 over the phone's id and a 5-minute time step, keyed by a secret the phone and server already share. Codes are valid for 30 minutes, a code (or

any older one) can never be reused, and **5 wrong tries trigger a 10-minute wait**, so the code cannot be guessed.

8.4 Device hardening (Device Owner)

PROTECTION	EFFECT
Uninstall blocked	The customer cannot remove MoneyMobile.
Clear data / Force stop disabled	The app's saved state and service cannot be wiped or stopped from Settings.
Factory reset blocked in Settings	The normal reset option is unavailable while the phone is managed.
Safe mode, extra users, unknown sources blocked	No guest profile without the lock, no safe-mode escape, no sideloaded tools.
Automatic date and time forced	Together with the trusted clock, moving the date back does not delay a lock.
True kiosk (Lock Task)	While locked: no Home, Recents, notification shade or power menu. Calls and SMS to the shop still work. The lock screen is the Home app, so it returns after a reboot.
USB locked while locked	No file transfer or USB debugging while the phone is locked; restored in full on unlock.
Airplane mode blocked while locked; Wi-Fi kept on	A locked phone always stays reachable so the shop's unlock can arrive.
Bootloader state reported	Phones with an unlocked bootloader are flagged "high risk" on the dashboard.
Per-shop reset protection	After any factory reset, the phone only sets up with the shop's Google account. The list is signed, and removal at release is verified before control is given up.

8.5 App update integrity

Only the super admin can publish. The server reads the package name and version from inside the uploaded APK, rejects other apps and non-newer builds, and stores its SHA-256. The phone accepts only a **signed** offer over **HTTPS**, checks the file's **size and SHA-256**, and verifies **package name, exact version and signing certificate** before installing. Installs are atomic, the "unknown sources" restriction is lifted only during the install and restored on every start, and staged rollout (test in one shop → publish → withdraw) limits exposure.

8.6 Data protection

- **Encrypted storage on the phone** (EncryptedSharedPreferences, AES-256). The lock flag is also kept in a plain shadow file that fails **safe**: if encrypted storage cannot open, the phone stays locked.
- **Release build hardening:** not debuggable, no network logging, HTTPS only (cleartext disabled), and app backup disabled.
- **Private file storage:** NIC photos and app files live in private buckets and are reached only through short-lived signed links.

- **Secrets stay out of source control:** environment files, the Firebase key, keystores and per-shop setup configs are git-ignored.

8.7 Business-rule integrity

- All money math (interest, schedules, payment allocation, balances) runs **on the server**. Numbers from the browser are never trusted.
- Each lock carries a reason: `overdue`, `sim` or `manual`. A payment can only lift an overdue lock.
- License limits are enforced twice: in the server action and by a database trigger.
- Release is only possible at a zero balance, and the server checks this again.

9 · RELIABILITY

Built to keep working

Phones lose signal, run out of data, reboot and get updated. Servers have hiccups. MoneyMobile is designed so that none of these leave a phone in the wrong state.

Delivery is guaranteed, speed is a bonus

- Check-in every ~60 seconds while the service runs.
- Immediate check-in when any network comes up.
- Firebase push wakes the phone within seconds.
- WorkManager fallback every 4 hours if the service stops.
- An unconfirmed command is handed over again after 90 seconds.

Offline is a normal state

- Overdue lock works with no internet, on a trusted clock.
- Each month's deadline is stored ahead of time.
- Reminders 48 h and 24 h before, shown even after a reboot.
- SIM removal locks instantly; the alert is sent when online.
- Unlock codes are checked on the phone itself.

Survives restarts and updates

- Boot receiver restores the lock and re-arms alarms.
- After an app update the new version restarts management at once.
- Sticky foreground service, restarted if killed.
- Lock screen set as Home while locked, so it is always on top.

Safe failure on the server

- A database error returns "try later" (503), never "unknown phone" (401).
- Missing migrations fall back instead of breaking pages.
- If the reset-protection list can't be read, nothing is sent, so phones keep their protection.
- Heartbeat database work runs in parallel for fast replies.

Consistency rules that prevent surprises

- The phone never unlocks on missing data: it only acts on an explicit, verified "locked: false".
- While a SIM alert is still undelivered, the server's "unlocked" is not applied, so the SIM lock cannot be undone by accident.
- The phone waits a moment past each deadline so phone and server never disagree for a few seconds.
- Acknowledgements retry with growing pauses; Release keeps trying for about a minute on a weak connection.
- A phone whose record was deleted registers itself again using its saved shop setup.

10 · USABILITY

How each user experiences MoneyMobile

The guiding idea: nobody should need training, a manual or technical words to use it.

Shop owner: in control, at a glance

Totals for financed, collected and outstanding money. Every phone's status in one list: online or offline, app version, reset protection, risk flags. Live command status that updates on its own. Clear limits and plain error messages, for example "License limit reached — please contact the super admin."

Employee: fast, guided daily work

Set up a phone with one double-click. Record a payment and watch the phone unlock in seconds. Generate an unlock code while the customer is on the line. Every button shows its progress ("Sending...", "Waiting for the phone to check in...") and can't be double-clicked by mistake.

Customer: informed and treated fairly

The app shows the balance, next due date and the exact lock date ("in 4 days 3 h"). Reminders arrive 2 days and 1 day before. If the phone is locked, the screen explains why and offers Call shop, SMS shop, Wi-Fi, Enter code and Request unlock. Calls keep working.

Super admin: one dashboard for the platform

Every shop, owner and license in one place. Publishing an app update takes three clicks (upload, test, publish) and also refreshes every shop's setup kit. Shops that haven't set up reset protection are flagged.

Usability principles applied

PRINCIPLE	EXAMPLE IN MONEYMOBILE
Plain language	"Waiting for the phone to check in", "Paused on the phone — connection dropped, continues automatically", not error codes.
Visible system status	Command states (queued → delivered → applied), update progress and protection badges refresh every few seconds while something is in progress.
Error prevention	Release is disabled until the balance is zero and needs a confirmation. Buttons disable while sending. Invalid account IDs are explained, not just rejected.
One time zone everywhere	All dates and times are shown in Sri Lanka time, identical on server and browser.
Responsive	Works on desktop and mobile, with a navigation drawer on small screens.
Accessible motion	Animations are switched off for people who prefer reduced motion.

11 · APPLE DESIGN PRINCIPLES

Designed with Apple's Human Interface Guidelines

Both the website and the Android app follow Apple's design principles: clarity, deference and depth. Wording follows the Apple Style Guide. The aim is a calm, trustworthy interface that does not look generated or generic.

Clarity

Large, tight titles, strong text contrast, sentence-case labels and one clear primary action per screen.

Deference

Quiet grays and white cards let the content (balances, statuses, customers) lead. No gradients, no emoji, no decoration for its own sake.

Depth

Soft shadows, rounded grouped cards and layered sheets show hierarchy without heavy borders.

The design system

ELEMENT	IMPLEMENTATION
Typography	Apple's system font stack (`-apple-system` , SF Pro Text / Display) with Inter as fallback, tighter letter-spacing on headings, kerning and ligatures on.
Colour	System blue (actions) Green (paid, online) Orange (warnings) Red (locked, overdue) Grouped background
Semantic tokens	Colours are named by meaning (background, label, secondary label, card, destructive) and defined once, so every screen stays consistent.
Shape	Continuous rounded corners (about 14 px), inset grouped lists and cards like iOS Settings.
Interaction	Gentle press feedback (buttons scale to 98%), quiet rounded focus rings, thin scrollbars.
Android app	Apple system colours as Android resources, light and dark themes that follow the phone, inset grouped lists, large titles, flat cards and system blue, green, red and orange.
Status badges	Small rounded pills in tinted system colours: online updating locked published

> The result is a product that feels familiar to anyone who has used an iPhone or Mac, while running on Android phones and any web browser.

12 · MARKET POSITION AND STRENGTHS

Where MoneyMobile stands out

Installment shops usually combine three things: a ledger (often paper or spreadsheets), a general device-management tool, and trust in the customer. MoneyMobile replaces all three with one purpose-built system.

CAPABILITY	MONEYMOBILE	GENERAL MDM TOOLS	LEDGER / SPREADSHEET
Installment plans, schedules, payments	Built in	Not their purpose	Manual
Lock driven by payment status	Automatic, monthly	Not their purpose	No
Lock with no internet on the due date	Yes, trusted clock	Typically needs a connection	No
Unlock the moment the customer pays	Yes, by push	Manual	No
Offline unlock code by phone call	Yes, 30 minutes	Varies	No
SIM removal lock (dual-SIM)	Yes	Varies	No
Customer-facing balance and reminders	Yes	No	No
Per-shop reset protection	Yes	Organisation-level	No
Multi-shop with licenses	Yes	Organisation-based	No
Local fit: LKR, Sri Lanka time, low-data use	Yes	Generic	—

Google's own Android Management API policy does not permit device-financing or payment-based locking, which keeps general enterprise MDM tools built on it out of this use case.

Our strongest areas

1 · Enforcement that does not depend on the network

Server-signed monthly deadlines, a clock the customer cannot move, and on-phone unlock codes mean the agreement is enforced even on a WhatsApp-only data package.

2 · Cryptographic trust between server and phone

Per-phone signing keys, hashed device tokens, replay guards and verified updates. A fake server or forged message cannot unlock, release or update a phone.

3 · Fair, transparent rules

Grace periods, reminders, calls kept open, lock reasons kept separate, instant unlock on payment, and a verified clean release when the phone is paid off.

4 · Operational simplicity

One-click setup kits, QR provisioning, one-click online updates, staged rollout, and a single dashboard for every shop on the platform.

13 · TECHNOLOGY AND CODEBASE

Stack and structure

151
TypeScript files, about 8,400 lines (web)

35
Kotlin files, about 4,900 lines (Android)

21
database migrations, each safe to run more than once

LAYER	TECHNOLOGY
Web framework	Next.js 14 (App Router, Server Components, Server Actions), React 18, TypeScript 5 (strict type-checking on every change)
Interface	Tailwind CSS with an Apple-style token system, lucide icons, class-variance-authority components
Forms and validation	React server actions with progressive forms; every input is validated and re-checked on the server
Database and auth	Supabase: PostgreSQL with Row-Level Security, Supabase Auth (JWT sessions), Supabase Storage (private buckets)
Hosting	Vercel (serverless functions); Supabase cloud database
Push messaging	Firebase Cloud Messaging (HTTP v1, service-account auth)
Android app	Kotlin, Android 8.0+ (API 26), target SDK 34, Device Owner APIs, Lock Task (kiosk), WorkManager, Coroutines, Retrofit + OkHttp + Gson, AndroidX Security (encrypted storage), Firebase Messaging
Cryptography	HMAC-SHA256 (signatures, unlock codes), SHA-256 (token hashes, file checks), AES-256 (on-device storage), TLS (all traffic in release builds)
Setup tooling	One-click setup scripts for macOS and Windows (ADB), QR-code provisioning, per-shop generated configuration

Device API endpoints

ENDPOINT	PURPOSE
`POST /api/devices/enroll`	Register a phone with a one-time enrollment token.
`POST /api/devices/auto-enroll`	Register a phone with the shop's secret (USB kit or QR).
`POST /api/devices/heartbeat`	Check in: report status; receive signed lock state, deadlines, command, update offer, reset-protection list.
`POST /api/devices/command-ack`	Confirm a command was applied (Release marks the phone handed over).
`POST /api/devices/security-event`	Report SIM removal, offline unlock-code use and tamper events.
`POST /api/devices/unlock-request`	Customer asks the shop for an unlock from the lock screen.
`POST /api/devices/app-update`	Report online-update progress: downloading, waiting, installing, installed, failed.

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